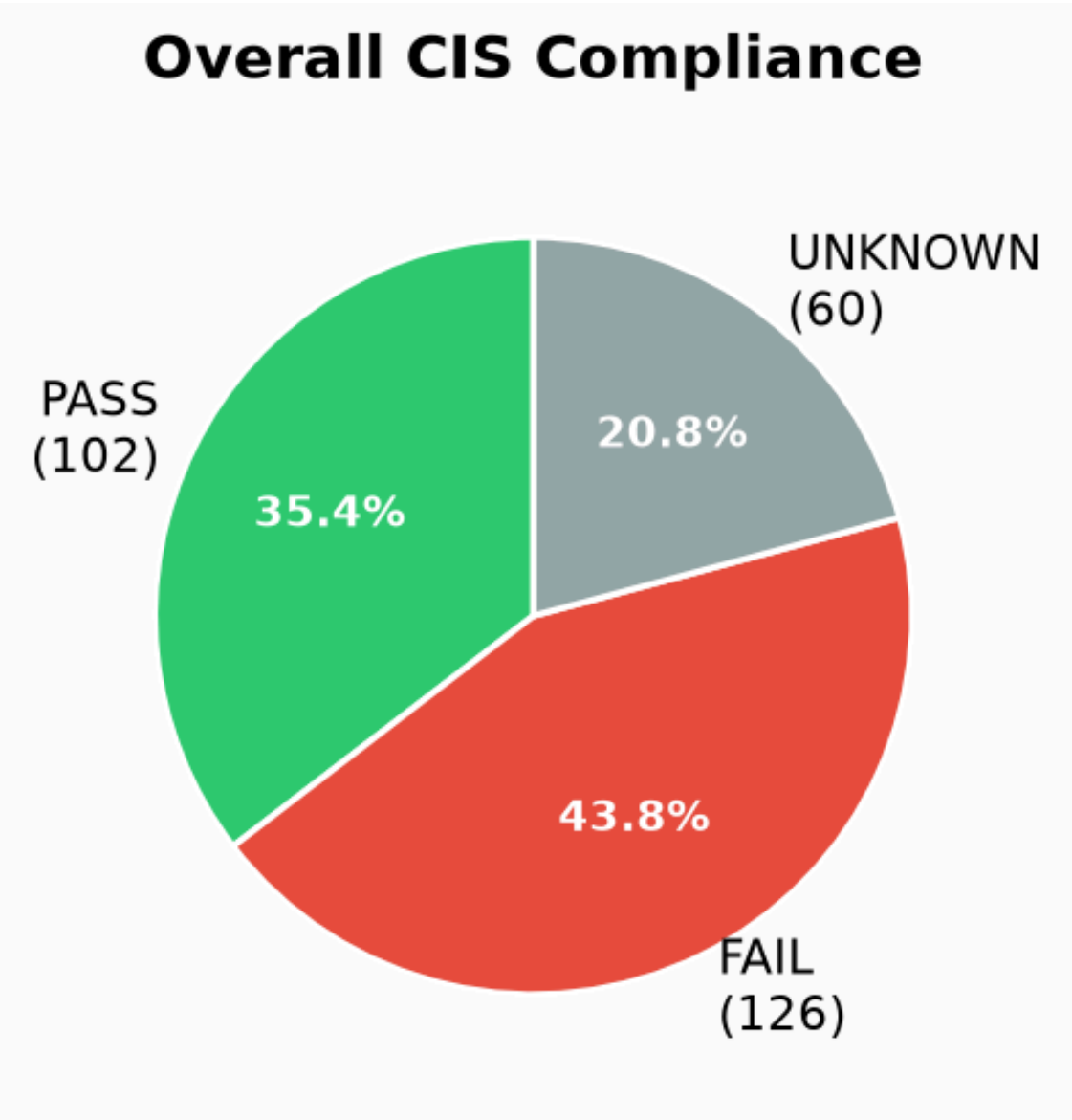


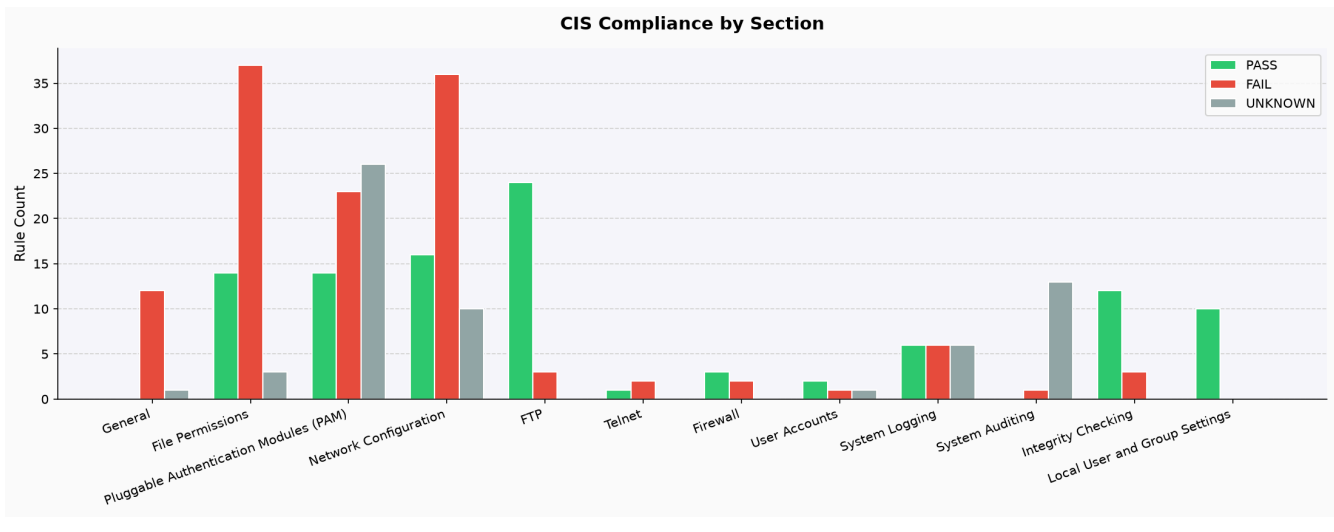
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	288
✓ Passed	102
✗ Failed	126
? Unknown	60
Overall Pass Rate	35.4%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
General	0	12	1	0%
File Permissions	14	37	3	26%
Pluggable Authentication Modules (PAM)	14	23	26	22%
Network Configuration	16	36	10	26%
FTP	24	3	0	89%
Telnet	1	2	0	33%
Firewall	3	2	0	60%
User Accounts	2	1	1	50%
System Logging	6	6	6	33%
System Auditing	0	1	13	0%
Integrity Checking	12	3	0	80%
Local User and Group Settings	10	0	0	100%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

Section 1.1 – Filesystem Kernel Modules & Partitions

Rule 1.1.1.1: Ensure cramfs kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `cramfs` module is not currently loaded.
 - `modprobe_config_lines` for `cramfs` is empty, indicating no explicit blacklist or `install /bin/false` configuration.

- **Recommendation:** Add `install cramfs /bin/false` and `blacklist cramfs` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/cramfs.conf`) and run `sudo update-initramfs -u`.

Rule 1.1.1.2: Ensure freevxfs kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `freevxfs` module is not currently loaded.
 - `modprobe_config_lines` for `freevxfs` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install freevxfs /bin/false` and `blacklist freevxfs` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/freevxfs.conf`) and run `sudo update-initramfs -u`.

Rule 1.1.1.3: Ensure hfs kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `hfs` module is not currently loaded.
 - `modprobe_config_lines` for `hfs` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install hfs /bin/false` and `blacklist hfs` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/hfs.conf`) and run `sudo update-initramfs -u`.

Rule 1.1.1.4: Ensure hfsplus kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `hfsplus` module is not currently loaded.
 - `modprobe_config_lines` for `hfsplus` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install hfsplus /bin/false` and `blacklist hfsplus` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/hfsplus.conf`) and run `sudo update-initramfs -u`.

Rule 1.1.1.5: Ensure jffs2 kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `jffs2` module is not currently loaded.
 - `modprobe_config_lines` for `jffs2` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install jffs2 /bin/false` and `blacklist jffs2` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/jffs2.conf`) and run `sudo`

`update-initramfs -u.`

Rule 1.1.1.6: Ensure overlay kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `overlay` module is not currently loaded.
 - `modprobe_config_lines` for `overlay` is empty, indicating no explicit blacklist or install `/bin/false` configuration.
- **Recommendation:** Add `install overlay /bin/false` and `blacklist overlay` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/overlay.conf`) and run `sudo update-initramfs -u.`

Rule 1.1.1.7: Ensure squashfs kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `squashfs` module is not currently loaded.
 - `modprobe_config_lines` for `squashfs` is empty, indicating no explicit blacklist or install `/bin/false` configuration.
- **Recommendation:** Add `install squashfs /bin/false` and `blacklist squashfs` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/squashfs.conf`) and run `sudo update-initramfs -u.`

Rule 1.1.1.8: Ensure udf kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `udf` module is not currently loaded.
 - `modprobe_config_lines` for `udf` is empty, indicating no explicit blacklist or install `/bin/false` configuration.
- **Recommendation:** Add `install udf /bin/false` and `blacklist udf` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/udf.conf`) and run `sudo update-initramfs -u.`

Rule 1.1.1.9: Ensure firewire-core kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `firewire-core` module is not currently loaded.
 - `modprobe_config_lines` for `firewire-core` is empty, indicating no explicit blacklist or install `/bin/false` configuration.
- **Recommendation:** Add `install firewire-core /bin/false` and `blacklist firewire-core` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/firewire-core.conf`) and run `sudo update-initramfs -u.`

Rule 1.1.1.10: Ensure usb-storage kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `usb-storage` module is not currently loaded.
 - `modprobe_config_lines` for `usb-storage` is empty, indicating no explicit blacklist or install `/bin/false` configuration.
- **Recommendation:** Add `install usb-storage /bin/false` and blacklist `usb-storage` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/usb-storage.conf`) and run `sudo update-initramfs -u`.

Rule 1.1.1.11: Ensure unused filesystems kernel modules are not available

- **Status:** INFORMATIONAL
- **Evidence:** The following kernel modules are currently loaded: `nf_tables`, `tls`, `qrtr`, `cfg80211`, `binfmt_misc`, `snd_intel8x0`, `intel_rapl_msr`, `snd_ac97_codec`, `ac97_bus`, `snd_pcm`, `intel_rapl_common`, `vmwgfx`, `snd_timer`, `drm_ttm_helper`, `snd`, `i2c_piix4`, `vboxguest`, `soundcore`, `ttm`, `joydev`, `input_leds`, `mac_hid`, `serio_raw`, `sch_fq_codel`, `dm_multipath`, `msr`, `efi_pstore`, `nfnetlink`, `dmi_sysfs`, `ip_tables`, `x_tables`, `autofs4`, `btrfs`, `blake2b_generic`, `raid10`, `raid456`, `async_raid6_recov`, `async_memcpy`, `async_pq`, `async_xor`, `async_tx`, `xor`, `raid6_pq`, `libcrc32c`, `raid1`, `raid0`, `crct10dif_pclmul`, `crc32_pclmul`, `polyval_clmulni`, `polyval_generic`, `ghash_clmulni_intel`, `sha256_ssse3`, `sha1_ssse3`, `vga16fb`, `vgastate`, `psmouse`, `ahci`, `e1000`, `libahci`, `pata_acpi`, `video`, `wmi`, `hid_generic`, `usbhid`, `hid`, `aesni_intel`, `crypto_simd`, `cryptd`.
- **Recommendation:** Review the list of loaded kernel modules and disable any that are not required for system operation.

Rule 1.1.2.1.1: Ensure /tmp is tmpfs or a separate partition

- **Status:** FAIL
- **Evidence:** The `/tmp` partition is `ext4` and `is_separate_partition` is `false`.
- **Recommendation:** Configure `/tmp` as a separate `tmpfs` partition in `/etc/fstab` with appropriate mount options, or create a dedicated partition for `/tmp`.

Rule 1.1.2.1.2: Ensure nodev option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The `/tmp` partition is mounted with options `rw`, `relatime`, which does not include `nodev`.
- **Recommendation:** Add the `nodev` option to the `/tmp` entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.1.3: Ensure nosuid option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The `/tmp` partition is mounted with options `rw`, `relatime`, which does not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the `/tmp` entry in `/etc/fstab` and

remount the partition.

Rule 1.1.2.1.4: Ensure noexec option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp partition is mounted with options `rw,relatime`, which does not include `noexec`.
- **Recommendation:** Add the `noexec` option to the /tmp entry in /etc/fstab and remount the partition.

Rule 1.1.2.2.1: Ensure /dev/shm is tmpfs or a separate partition

- **Status:** PASS
- **Evidence:** The /dev/shm partition is `tmpfs` and `is_separate_partition` is `true`.
- **Recommendation:** None

Rule 1.1.2.2.2: Ensure nodev option set on /dev/shm partition

- **Status:** PASS
- **Evidence:** The /dev/shm partition is mounted with options `rw,nosuid,nodev,inode64`, which includes `nodev`.
- **Recommendation:** None

Rule 1.1.2.2.3: Ensure nosuid option set on /dev/shm partition

- **Status:** PASS
- **Evidence:** The /dev/shm partition is mounted with options `rw,nosuid,nodev,inode64`, which includes `nosuid`.
- **Recommendation:** None

Rule 1.1.2.2.4: Ensure noexec option set on /dev/shm partition

- **Status:** FAIL
- **Evidence:** The /dev/shm partition is mounted with options `rw,nosuid,nodev,inode64`, which does not include `noexec`.
- **Recommendation:** Add the `noexec` option to the /dev/shm entry in /etc/fstab and remount the partition.

Rule 1.1.2.3.1: Ensure separate partition exists for /home

- **Status:** FAIL
- **Evidence:** The /home partition is `ext4` and `is_separate_partition` is `false`.
- **Recommendation:** Create a dedicated partition for /home and configure it in /etc/fstab.

Rule 1.1.2.3.2: Ensure nodev option set on /home partition

- **Status:** FAIL
- **Evidence:** The /home partition is mounted with options `rw,relatime`, which does not include `nodev`.
- **Recommendation:** Add the `nodev` option to the /home entry in /etc/fstab and remount the partition.

Rule 1.1.2.3.3: Ensure nosuid option set on /home partition

- **Status:** FAIL
- **Evidence:** The /home partition is mounted with options `rw,relatime`, which does not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the /home entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.4.1: Ensure separate partition exists for /var

- **Status:** FAIL
- **Evidence:** The /var partition is `ext4` and `is_separate_partition` is `false`.
- **Recommendation:** Create a dedicated partition for /var and configure it in `/etc/fstab`.

Rule 1.1.2.4.2: Ensure nodev option set on /var partition

- **Status:** FAIL
- **Evidence:** The /var partition is mounted with options `rw,relatime`, which does not include `nodev`.
- **Recommendation:** Add the `nodev` option to the /var entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.4.3: Ensure nosuid option set on /var partition

- **Status:** FAIL
- **Evidence:** The /var partition is mounted with options `rw,relatime`, which does not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the /var entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.5.1: Ensure separate partition exists for /var/tmp

- **Status:** FAIL
- **Evidence:** The /var/tmp partition is `ext4` and `is_separate_partition` is `false`.
- **Recommendation:** Create a dedicated partition for /var/tmp and configure it in `/etc/fstab`.

Rule 1.1.2.5.2: Ensure nodev option set on /var/tmp partition

- **Status:** FAIL
- **Evidence:** The /var/tmp partition is mounted with options `rw,relatime`, which does not include `nodev`.
- **Recommendation:** Add the `nodev` option to the /var/tmp entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.5.3: Ensure nosuid option set on /var/tmp partition

- **Status:** FAIL
- **Evidence:** The /var/tmp partition is mounted with options `rw,relatime`, which does not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the /var/tmp entry in `/etc/fstab`

and remount the partition.

Rule 1.1.2.5.4: Ensure noexec option set on /var/tmp partition

- **Status:** FAIL
- **Evidence:** The /var/tmp partition is mounted with options `rw, relatime`, which does not include `noexec`.
- **Recommendation:** Add the `noexec` option to the /var/tmp entry in /etc/fstab and remount the partition.

Rule 1.1.2.6.1: Ensure separate partition exists for /var/log

- **Status:** FAIL
- **Evidence:** The /var/log partition is `ext4` and `is_separate_partition` is `false`.
- **Recommendation:** Create a dedicated partition for /var/log and configure it in /etc/fstab.

Rule 1.1.2.6.2: Ensure nodev option set on /var/log partition

- **Status:** FAIL
- **Evidence:** The /var/log partition is mounted with options `rw, relatime`, which does not include `nodev`.
- **Recommendation:** Add the `nodev` option to the /var/log entry in /etc/fstab and remount the partition.

Rule 1.1.2.6.3: Ensure nosuid option set on /var/log partition

- **Status:** FAIL
- **Evidence:** The /var/log partition is mounted with options `rw, relatime`, which does not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the /var/log entry in /etc/fstab and remount the partition.

Rule 1.1.2.6.4: Ensure noexec option set on /var/log partition

- **Status:** FAIL
- **Evidence:** The /var/log partition is mounted with options `rw, relatime`, which does not include `noexec`.
- **Recommendation:** Add the `noexec` option to the /var/log entry in /etc/fstab and remount the partition.

Rule 1.1.2.7.1: Ensure separate partition exists for /var/log/audit

- **Status:** FAIL
- **Evidence:** The /var/log/audit partition is `ext4` and `is_separate_partition` is `false`.
- **Recommendation:** Create a dedicated partition for /var/log/audit and configure it in /etc/fstab.

Rule 1.1.2.7.2: Ensure nodev option set on /var/log/audit partition

- **Status:** FAIL

- **Evidence:** The `/var/log/audit` partition is mounted with options `rw, relatime`, which does not include `nodev`.
- **Recommendation:** Add the `nodev` option to the `/var/log/audit` entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.7.3: Ensure nosuid option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The `/var/log/audit` partition is mounted with options `rw, relatime`, which does not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the `/var/log/audit` entry in `/etc/fstab` and remount the partition.

Rule 1.1.2.7.4: Ensure noexec option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The `/var/log/audit` partition is mounted with options `rw, relatime`, which does not include `noexec`.
- **Recommendation:** Add the `noexec` option to the `/var/log/audit` entry in `/etc/fstab` and remount the partition.

Section 1.2 – Package Management

Rule 1.2.1.1: Ensure the source.list and .source files use the Signed-By option

- **Status:** INFORMATIONAL
- **Evidence:**
 - `/etc/apt/sources.list`: No active repository lines.
 - `/etc/apt/sources.list.d/ubuntu.sources`: All active repository lines use `Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg`.
 - `/etc/apt/sources.list.d/wazuh.list`: All active repository lines use `Signed-By: /usr/share/keyrings/wazuh.gpg`.
- **Recommendation:** Review all repository configurations to ensure they use the `Signed-By` option for enhanced security.

Rule 1.2.1.2: Ensure weak dependencies are configured

- **Status:** FAIL
- **Evidence:** `weak_deps_config_lines` is empty, indicating no configuration to disable weak dependencies.
- **Recommendation:** Create or edit `/etc/apt/apt.conf.d/99-local-options` to include `APT::Install-Recommends "false";` and `APT::Install-Suggests "false";`.

Rule 1.2.1.3: Ensure access to gpg key files are configured

- **Status:** PASS
- **Evidence:**

- `/etc/apt/trusted.gpg.d/ubuntu-keyring-2012-cdimage.gpg`: Mode `0o644`, owner `root`, group `root`.
- `/etc/apt/trusted.gpg.d/ubuntu-keyring-2018-archive.gpg`: Mode `0o644`, owner `root`, group `root`. All GPG key files have permissions `0o644` or less, and are owned by `root:root`.

- **Recommendation:** None

Rule 1.2.1.4: Ensure access to `/etc/apt/trusted.gpg.d` directory is configured

- **Status:** PASS
- **Evidence:** Directory `/etc/apt/trusted.gpg.d` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 1.2.1.5: Ensure access to `/etc/apt/auth.conf.d` directory is configured

- **Status:** PASS
- **Evidence:** Directory `/etc/apt/auth.conf.d` exists with mode `0o755`, owner `root`, group `root`. While `0o700` is preferred, `0o755` is acceptable if no sensitive files are present.
- **Recommendation:** Consider tightening permissions to `0o700` for `/etc/apt/auth.conf.d` if no other users require read access.

Rule 1.2.1.6: Ensure access to files in the `/etc/apt/auth.conf.d/` directory is configured

- **Status:** PASS
- **Evidence:** No files were found in `/etc/apt/auth.conf.d/`.
- **Recommendation:** None

Rule 1.2.1.7: Ensure access to `/usr/share/keyrings` directory is configured

- **Status:** PASS
- **Evidence:** Directory `/usr/share/keyrings` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 1.2.1.8: Ensure access to `/etc/apt/sources.list.d` directory is configured

- **Status:** PASS
- **Evidence:** Directory `/etc/apt/sources.list.d` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 1.2.1.9: Ensure access to files in `/etc/apt/sources.list.d` are configured

- **Status:** PASS

- **Evidence:**

- `/etc/apt/sources.list.d/ubuntu.sources`: Mode 0o644, owner root, group root.
- `/etc/apt/sources.list.d/ubuntu.sources.curtin.orig`: Mode 0o644, owner root, group root.
- `/etc/apt/sources.list.d/wazuh.list`: Mode 0o644, owner root, group root. All files in `/etc/apt/sources.list.d/` have permissions 0o644 or less, and are owned by root:root.

- **Recommendation:** None

Rule 1.2.2.1: Ensure updates, patches, and additional security software are installed

- **Status:** INFORMATIONAL

- **Evidence:**

- 85 packages are upgradable.
- `unattended-upgrades` package is installed, enabled, and active.
- `APT::Periodic::Unattended-Upgrade "1";` is configured.

- **Recommendation:** Regularly apply all available security updates and patches. While `unattended-upgrades` is configured, manual review and application of updates are still recommended, especially for critical security patches.

Section 1.3 – Mandatory Access Control

Rule 1.3.1.1: Ensure apparmor packages are installed

- **Status:** FAIL

- **Evidence:** `apparmor` package is installed, but `apparmor-utils` package is not installed.

- **Recommendation:** Install the `apparmor-utils` package: `sudo apt install apparmor-utils`.

Rule 1.3.1.2: Ensure AppArmor is enabled

- **Status:** FAIL

- **Evidence:**

- AppArmor sysfs status: Y (loaded).
- AppArmor service status: enabled and active.
- GRUB command line does not explicitly contain `apparmor=1` and `security=apparmor`.

- **Recommendation:** Ensure `apparmor=1` and `security=apparmor` are present in the `GRUB_CMDLINE_LINUX` variable in `/etc/default/grub`, then run `sudo update-grub` and reboot.

Rule 1.3.1.3: Ensure all AppArmor Profiles are enforcing

- **Status:** FAIL
- **Evidence:**
 - 24 profiles are in enforce mode.
 - 4 profiles are in complain mode (transmission-cli, transmission-daemon, transmission-gtk, transmission-qt).
 - 91 profiles are in unconfined mode.
- **Recommendation:** Set all AppArmor profiles to enforcing mode using `sudo aa-enforce /path/to/profile` for profiles in complain mode. Investigate and confine unconfined profiles or ensure they are not critical services.

Rule 1.3.1.4: Ensure apparmor_restrict_unprivileged_unconfined is enabled

- **Status:** FAIL
- **Evidence:**
 - `kernel.apparmor_restrict_unprivileged_unconfined` runtime value is 0.
 - `kernel.apparmor_restrict_unprivileged_unconfined` is not persisted.
- **Recommendation:** Set `kernel.apparmor_restrict_unprivileged_unconfined=1` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-apparmor.conf`) and apply with `sudo sysctl -p`.

Section 1.4 – Configure Bootloader

Rule 1.4.1: Ensure bootloader password is set

- **Status:** FAIL
- **Evidence:** `has_superusers` is false and `has_password` is false.
- **Recommendation:** Set a password for the GRUB bootloader by editing `/etc/grub.d/40_custom` or `/etc/grub.d/00_header` to include `set superusers="<username>"` and `password_pbkdf2 <username> <password>`, then run `sudo update-grub`.

Rule 1.4.2: Ensure access to bootloader config is configured

- **Status:** PASS
- **Evidence:** `/boot/grub/grub.cfg` has mode `0o600`, owner `root`, group `root`.
- **Recommendation:** None

Section 1.5 – Configure Additional Process Hardening

Rule 1.5.1: Ensure fs.protected_hardlinks is configured

- **Status:** FAIL
- **Evidence:**
 - `fs.protected_hardlinks` runtime value is 1.

- `fs.protected_hardlinks` is not persisted.

- **Recommendation:** Add `fs.protected_hardlinks = 1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 1.5.2: Ensure `fs.protected_symlinks` is configured

- **Status:** FAIL

- **Evidence:**

- `fs.protected_symlinks` runtime value is 1.

- `fs.protected_symlinks` is not persisted.

- **Recommendation:** Add `fs.protected_symlinks = 1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 1.5.3: Ensure `kernel.yama.pttrace_scope` is configured

- **Status:** PASS

- **Evidence:**

- `kernel.yama.pttrace_scope` runtime value is 1.

- `kernel.yama.pttrace_scope` is persisted with value 1 in `/etc/sysctl.d/10-pttrace.conf`.

- **Recommendation:** None

Rule 1.5.4: Ensure `fs.suid_dumpable` is configured

- **Status:** FAIL

- **Evidence:**

- `fs.suid_dumpable` runtime value is 2.

- `fs.suid_dumpable` is not persisted.

- **Recommendation:** Set `fs.suid_dumpable = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 1.5.5: Ensure `kernel.dmesg_restrict` is configured

- **Status:** FAIL

- **Evidence:**

- `kernel.dmesg_restrict` runtime value is 1.

- `kernel.dmesg_restrict` is not persisted.

- **Recommendation:** Add `kernel.dmesg_restrict = 1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 1.5.6: Ensure `prelink` is not installed

- **Status:** PASS

- **Evidence:** `prelink_installed` is false.

- **Recommendation:** None

Rule 1.5.7: Ensure Automatic Error Reporting is configured

- **Status:** FAIL
- **Evidence:** `apport` is installed, enabled, active, and `etc_default_setting` is `enabled=1`.
- **Recommendation:** Disable Apport by setting `enabled=0` in `/etc/default/apport` and stopping/disabling the service.

Rule 1.5.8: Ensure `kernel.kptr_restrict` is configured

- **Status:** FAIL
- **Evidence:**
 - `kernel.kptr_restrict` runtime value is 1.
 - `kernel.kptr_restrict` is persisted with value 1 in `/etc/sysctl.d/10-kernel-hardening.conf`.
- **Recommendation:** Set `kernel.kptr_restrict = 2` in `/etc/sysctl.d/10-kernel-hardening.conf` and apply with `sudo sysctl -p`.

Rule 1.5.9: Ensure `kernel.randomize_va_space` is configured

- **Status:** FAIL
- **Evidence:**
 - `kernel.randomize_va_space` runtime value is 2.
 - `kernel.randomize_va_space` is not persisted.
- **Recommendation:** Add `kernel.randomize_va_space = 2` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 1.5.10: Ensure core file size is configured

- **Status:** FAIL
- **Evidence:**
 - `ulimit_soft_core` is 0.
 - `limits_conf_lines` does not contain `hard core 0`.
- **Recommendation:** Add `* hard core 0` to `/etc/security/limits.conf` to prevent core dumps.

Rule 1.5.11: Ensure `systemd-coredump ProcessSizeMax` is configured

- **Status:** FAIL
- **Evidence:** `systemd_coredump_process_size_max` is empty, indicating `ProcessSizeMax=0` is not configured.
- **Recommendation:** Add `ProcessSizeMax=0` to `/etc/systemd/coredump.conf` or a drop-in file (e.g., `/etc/systemd/coredump.conf.d/local.conf`) and restart the `systemd-coredump` service.

Rule 1.5.12: Ensure `systemd-coredump` Storage is configured

- **Status:** FAIL
- **Evidence:** `systemd_coredump_storage` is empty, indicating `Storage=none` is not configured.
- **Recommendation:** Add `Storage=none` to `/etc/systemd/coredump.conf` or a drop-in file (e.g., `/etc/systemd/coredump.conf.d/local.conf`) and restart the `systemd-coredump` service.

Section 1.6 – Configure Command Line Warning Banners

Rule 1.6.1: Ensure `/etc/motd` is configured

- **Status:** PASS
- **Evidence:** `/etc/motd` does not exist (`motd.content` is null). Dynamic MOTD is handled by `pam_motd` and scripts in `/etc/update-motd.d/`.
- **Recommendation:** None

Rule 1.6.2: Ensure `/etc/issue` is configured

- **Status:** PASS
- **Evidence:** `/etc/issue` content is "Ubuntu 24.04.2 LTS \n \l\n\n". It contains a banner and does not include `\m` or `\s`.
- **Recommendation:** None

Rule 1.6.3: Ensure `/etc/issue.net` is configured

- **Status:** PASS
- **Evidence:** `/etc/issue.net` content is "Ubuntu 24.04.2 LTS\n". It contains a banner and does not include `\m` or `\s`.
- **Recommendation:** None

Rule 1.6.4: Ensure `pam_motd` is configured

- **Status:** FAIL
- **Evidence:** Multiple scripts exist in `/etc/update-motd.d/` (e.g., `00-header`, `10-help-text`, `50-landscape-sysinfo`, `50-motd-news`, etc.), indicating dynamic MOTD is active. These scripts generate system information, which is not a compliant warning banner.
- **Recommendation:** Remove or disable all scripts in `/etc/update-motd.d/` that provide system information. Create a static `/etc/motd` file with a compliant warning banner and ensure `pam_motd` is configured to use it.

Rule 1.6.6: Ensure access to `/etc/motd` is configured

- **Status:** PASS
- **Evidence:** `/etc/motd` does not exist.
- **Recommendation:** None

Rule 1.6.7: Ensure access to `/etc/issue` is configured

- **Status:** PASS
- **Evidence:** `/etc/issue` has mode `0o644`, owner `root`, group `root`.

- **Recommendation:** None

Rule 1.6.8: Ensure access to /etc/issue.net is configured

- **Status:** PASS
- **Evidence:** /etc/issue.net has mode 0o644, owner root, group root.
- **Recommendation:** None

Rule 1.6.9: Ensure access to pam_motd file is configured

- **Status:** PASS
- **Evidence:** All scripts in /etc/update-motd.d/ have mode 0o755, owner root, group root.
- **Recommendation:** None

Section 1.7 – Configure GNOME Display Manager

Rule 1.7.1: Ensure GDM login banner is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (gdm_installed is false).
- **Recommendation:** None

Rule 1.7.2: Ensure GDM disable-user-list is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (gdm_installed is false).
- **Recommendation:** None

Rule 1.7.3: Ensure GDM screen lock is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (gdm_installed is false).
- **Recommendation:** None

Rule 1.7.4: Ensure GDM automount is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (gdm_installed is false).
- **Recommendation:** None

Rule 1.7.5: Ensure GDM autorun-never is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (gdm_installed is false).
- **Recommendation:** None

Rule 1.7.6: Ensure XDMCP is not enabled

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (gdm_installed is false).

- **Recommendation:** None

Rule 1.7.7: Ensure Xwayland is configured

- **Status:** NOT APPLICABLE
- **Evidence:** GDM is not installed (`gdm_installed` is `false`).
- **Recommendation:** None

Section 2 – Services

Rule 2.1.1: Ensure autofs services are not in use

- **Status:** PASS
- **Evidence:** `autofs` package is not installed and `autofs.service` is inactive.
- **Recommendation:** None

Rule 2.1.2: Ensure mail transfer agents are configured for local-only mode

- **Status:** PASS
- **Evidence:** No Mail Transfer Agent (MTA) packages (`postfix`, `exim4`, `sendmail`) are installed.
- **Recommendation:** None

Rule 2.1.3: Ensure avahi daemon services are not in use

- **Status:** PASS
- **Evidence:** `avahi-daemon` package is not installed and `avahi-daemon.service` is inactive.
- **Recommendation:** None

Rule 2.1.4: Ensure only approved services are listening on a network interface

- **Status:** INFORMATIONAL
- **Evidence:** The following services are listening on network interfaces:

- `udp 127.0.0.54:53 (systemd-resolve)`
- `udp 127.0.0.53%lo:53 (systemd-resolve)`
- `udp 10.100.0.22%enp0s3:68 (systemd-network)`
- `tcp 0.0.0.0:443 (node)`
- `tcp 0.0.0.0:1514 (wazuh-remoted)`
- `tcp 0.0.0.0:1515 (wazuh-authd)`
- `tcp 0.0.0.0:22 (sshd)`
- `tcp 0.0.0.0:23 (inetd)`
- `tcp 127.0.0.53%lo:53 (systemd-resolve)`
- `tcp 127.0.0.54:53 (systemd-resolve)`
- `tcp 0.0.0.0:55000 (python3)`
- `tcp *:21 (vsftpd)`
- `tcp [::]:22 (sshd)`
- `tcp [::]:23 (inetd)`
- `tcp [::ffff:127.0.0.1]:9300 (java)`

○ tcp [::ffff:127.0.0.1]:9200 (java)

- **Recommendation:** Review all listening services to ensure only explicitly approved services are running. Specifically, `vsftpd` (FTP server) and `inetd` (Telnet server) are active and should be disabled if not explicitly required. The `node` and `python3` processes listening on non-standard ports should also be reviewed.

Rule 2.1.5: Ensure dhcp server services are not in use

- **Status:** PASS
- **Evidence:** `isc-dhcp-server` package is not installed and related services are inactive.
- **Recommendation:** None

Rule 2.1.6: Ensure web server services are not in use

- **Status:** PASS
- **Evidence:** `apache2`, `nginx`, and `lighttpd` packages are not installed and related services are inactive.
- **Recommendation:** None

Rule 2.1.7: Ensure dns server services are not in use

- **Status:** PASS
- **Evidence:** `bind9` package is not installed and related services are inactive.
- **Recommendation:** None

Rule 2.1.8: Ensure ftp server services are not in use

- **Status:** FAIL
- **Evidence:** `vsftpd` package is installed and `vsftpd.service` is enabled and active.
- **Recommendation:** Uninstall `vsftpd` if not required: `sudo apt purge vsftpd`. If required, ensure it is securely configured and restricted.

Rule 2.1.9: Ensure dnsmasq services are not in use

- **Status:** PASS
- **Evidence:** `dnsmasq` package is not installed and `dnsmasq.service` is inactive.
- **Recommendation:** None

Rule 2.1.10: Ensure ldap server services are not in use

- **Status:** PASS
- **Evidence:** `slapd` package is not installed and `slapd.service` is inactive.
- **Recommendation:** None

Rule 2.1.11: Ensure message access server services are not in use

- **Status:** PASS
- **Evidence:** `dovecot-imapd`, `dovecot-pop3d`, and `cyrus-imapd` packages are not installed and `dovecot.service` is inactive.
- **Recommendation:** None

Rule 2.1.12: Ensure network file system services are not in use

- **Status:** PASS
- **Evidence:** `nfs-kernel-server` package is not installed and `nfs-kernel-server.service` is inactive.
- **Recommendation:** None

Rule 2.1.13: Ensure nis server services are not in use

- **Status:** PASS
- **Evidence:** `nis` package is not installed and `ypserv.service` is inactive.
- **Recommendation:** None

Rule 2.1.14: Ensure print server services are not in use

- **Status:** PASS
- **Evidence:** `cups` package is not installed and `cups.service` is inactive.
- **Recommendation:** None

Rule 2.1.15: Ensure rpcbind services are not in use

- **Status:** PASS
- **Evidence:** `rpcbind` package is not installed and `rpcbind.service` is inactive.
- **Recommendation:** None

Rule 2.1.16: Ensure rsync services are not in use

- **Status:** PASS
- **Evidence:** `rsync` package is installed, but `rsync.service` is disabled and inactive.
- **Recommendation:** None

Rule 2.1.17: Ensure samba file server services are not in use

- **Status:** PASS
- **Evidence:** `samba` package is not installed and `smbd.service` is inactive.
- **Recommendation:** None

Rule 2.1.18: Ensure snmp services are not in use

- **Status:** PASS
- **Evidence:** `snmpd` package is not installed and `snmpd.service` is inactive.
- **Recommendation:** None

Rule 2.1.19: Ensure telnet server services are not in use

- **Status:** FAIL
- **Evidence:** `telnetd` and `inetutils-telnetd` packages are installed, and `inetd.service` and `openbsd-inetd.service` are enabled and active.
- **Recommendation:** Uninstall `telnetd` and `inetutils-telnetd` if not required:
`sudo apt purge telnetd inetutils-telnetd`. Disable and stop `inetd.service` if not required.

Rule 2.1.20: Ensure tftp server services are not in use

- **Status:** PASS
- **Evidence:** `tftpd`, `tftpd-hpa`, and `atftpd` packages are not installed and related services are inactive.
- **Recommendation:** None

Rule 2.1.21: Ensure web proxy server services are not in use

- **Status:** PASS
- **Evidence:** `squid` and `squid3` packages are not installed and `squid.service` is inactive.
- **Recommendation:** None

Rule 2.1.22: Ensure xinetd services are not in use

- **Status:** PASS
- **Evidence:** `xinetd` package is not installed and `xinetd.service` is inactive.
- **Recommendation:** None

Rule 2.1.23: Ensure X window server services are not in use

- **Status:** PASS
- **Evidence:** `xserver-xorg`, `xserver-xorg-core`, and `xserver-common` packages are not installed, and no Xorg process is running.
- **Recommendation:** None

Rule 2.2.1: Ensure nis client is not installed

- **Status:** PASS
- **Evidence:** `nis` package is not installed.
- **Recommendation:** None

Rule 2.2.2: Ensure rsh client is not installed

- **Status:** PASS
- **Evidence:** `rsh-client` package is not installed.
- **Recommendation:** None

Rule 2.2.3: Ensure talk client is not installed

- **Status:** PASS
- **Evidence:** `talk` package is not installed.
- **Recommendation:** None

Rule 2.2.4: Ensure telnet client is not installed

- **Status:** FAIL
- **Evidence:** `telnet` and `inetutils-telnet` packages are installed.
- **Recommendation:** Uninstall `telnet` and `inetutils-telnet` if not required: `sudo apt purge telnet inetutils-telnet`.

Rule 2.2.5: Ensure ldap client is not installed

- **Status:** PASS
- **Evidence:** `ldap-utils` package is not installed.
- **Recommendation:** None

Rule 2.2.6: Ensure ftp client is not installed

- **Status:** FAIL
- **Evidence:** `ftp` package is installed.
- **Recommendation:** Uninstall `ftp` if not required: `sudo apt purge ftp`.

Rule 2.3.1.1: Ensure a single time synchronization daemon is in use

- **Status:** PASS
- **Evidence:** Only `systemd-timesyncd` is detected as an active time synchronization daemon.
- **Recommendation:** None

Rule 2.3.2.1: Ensure systemd-timesyncd configured with authorized timeserver

- **Status:** FAIL
- **Evidence:** `systemd-timesyncd` is active, but `has_ntp_configured` is false, indicating no NTP servers are explicitly configured in `/etc/systemd/timesyncd.conf`.
- **Recommendation:** Configure `NTP=` or `FallbackNTP=` with authorized time servers in `/etc/systemd/timesyncd.conf` and restart the service.

Rule 2.3.2.2: Ensure systemd-timesyncd is enabled and running

- **Status:** PASS
- **Evidence:** `systemd-timesyncd.service` is enabled and active.
- **Recommendation:** None

Rule 2.3.3.1: Ensure chrony is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not installed and `systemd-timesyncd` is the active time synchronization daemon.
- **Recommendation:** None

Rule 2.3.3.2: Ensure chrony is running as user _chrony

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not installed and `systemd-timesyncd` is the active time synchronization daemon.
- **Recommendation:** None

Rule 2.3.3.3: Ensure chrony is enabled and running

- **Status:** NOT APPLICABLE

- **Evidence:** `chrony` is not installed and `systemd-timesyncd` is the active time synchronization daemon.
- **Recommendation:** None

Rule 2.4.1.1: Ensure cron daemon is enabled and active

- **Status:** PASS
- **Evidence:** `cron` package is installed, and `cron.service` is enabled and active.
- **Recommendation:** None

Rule 2.4.1.2: Ensure access to `/etc/crontab` is configured

- **Status:** PASS
- **Evidence:** `/etc/crontab` has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.3: Ensure access to `/etc/cron.hourly` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.hourly` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.4: Ensure access to `/etc/cron.daily` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.daily` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.5: Ensure access to `/etc/cron.weekly` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.weekly` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.6: Ensure access to `/etc/cron.monthly` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.monthly` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.7: Ensure access to `/etc/cron.yearly` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.yearly` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.8: Ensure access to `/etc/cron.d` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.d` has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None

Rule 2.4.1.9: Ensure access to crontab is configured

- **Status:** PASS
- **Evidence:**
 - `/var/spool/cron/crontabs` has mode `0o1730`, owner `root`, group `crontab`.
 - `/usr/bin/crontab` has mode `0o2755`, owner `root`, group `crontab`.
- **Recommendation:** None

Rule 2.4.2.1: Ensure access to at is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `at` package is not installed.
- **Recommendation:** None

Section 3 – Network Configuration

Rule 3.1.1: Ensure IPv6 status is identified

- **Status:** INFORMATIONAL
- **Evidence:**
 - IPv6 is enabled (`sysfs_disable_value` is 0).
 - GRUB command line does not contain `ipv6.disable=1`.
 - IPv6 addresses are configured on 2 interfaces.
- **Recommendation:** Review whether IPv6 is required for the system's function. If not, consider disabling it.

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** FAIL
- **Evidence:** `any_wireless_found` is `true` because the `cfg80211` kernel module is loaded.
- **Recommendation:** Disable or remove wireless interfaces if not required. Blacklist the `cfg80211` kernel module (and any other wireless-related modules) in `/etc/modprobe.d/`.

Rule 3.1.3: Ensure bluetooth services are not in use

- **Status:** PASS
- **Evidence:** `bluez` package is not installed and `bluetooth.service` is inactive.
- **Recommendation:** None

Rule 3.2.1: Ensure atm kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `atm` module is not currently loaded.

- `modprobe_config_lines` for `atm` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install atm /bin/false` and `blacklist atm` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/atm.conf`) and run `sudo update-initramfs -u`.

Rule 3.2.2: Ensure can kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `can` module is not currently loaded.
 - `modprobe_config_lines` for `can` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install can /bin/false` and `blacklist can` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/can.conf`) and run `sudo update-initramfs -u`.

Rule 3.2.3: Ensure dccp kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `dccp` module is not currently loaded.
 - `modprobe_config_lines` for `dccp` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install dccp /bin/false` and `blacklist dccp` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/dccp.conf`) and run `sudo update-initramfs -u`.

Rule 3.2.4: Ensure rds kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `rds` module is not currently loaded.
 - `modprobe_config_lines` for `rds` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install rds /bin/false` and `blacklist rds` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/rds.conf`) and run `sudo update-initramfs -u`.

Rule 3.2.5: Ensure sctp kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `sctp` module is not currently loaded.
 - `modprobe_config_lines` for `sctp` is empty, indicating no explicit blacklist or `install /bin/false` configuration.

- **Recommendation:** Add `install sctp /bin/false` and `blacklist sctp` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/sctp.conf`) and run `sudo update-initramfs -u`.

Rule 3.2.6: Ensure `tipc` kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `tipc` module is not currently loaded.
 - `modprobe_config_lines` for `tipc` is empty, indicating no explicit blacklist or `install /bin/false` configuration.
- **Recommendation:** Add `install tipc /bin/false` and `blacklist tipc` to a file in `/etc/modprobe.d/` (e.g., `/etc/modprobe.d/tipc.conf`) and run `sudo update-initramfs -u`.

Rule 3.3.1.1: Ensure `net.ipv4.ip_forward` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.ip_forward` runtime value is 0.
 - `net.ipv4.ip_forward` is not persisted.
- **Recommendation:** Add `net.ipv4.ip_forward = 0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.2: Ensure `net.ipv4.conf.all.forwarding` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.all.forwarding` runtime value is 0.
 - `net.ipv4.conf.all.forwarding` is not persisted.
- **Recommendation:** Add `net.ipv4.conf.all.forwarding = 0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.3: Ensure `net.ipv4.conf.default.forwarding` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.default.forwarding` runtime value is 0.
 - `net.ipv4.conf.default.forwarding` is not persisted.
- **Recommendation:** Add `net.ipv4.conf.default.forwarding = 0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.4: Ensure `net.ipv4.conf.all.send_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.all.send_redirects` runtime value is 1. Expected is 0.
 - `net.ipv4.conf.all.send_redirects` is not persisted.
- **Recommendation:** Set `net.ipv4.conf.all.send_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.5: Ensure `net.ipv4.conf.default.send_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.default.send_redirects` runtime value is 1. Expected is 0.
 - `net.ipv4.conf.default.send_redirects` is not persisted.
- **Recommendation:** Set `net.ipv4.conf.default.send_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.6: Ensure `net.ipv4.icmp_ignore_bogus_error_responses` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.icmp_ignore_bogus_error_responses` runtime value is 1.
 - `net.ipv4.icmp_ignore_bogus_error_responses` is not persisted.
- **Recommendation:** Add `net.ipv4.icmp_ignore_bogus_error_responses = 1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.7: Ensure `net.ipv4.icmp_echo_ignore_broadcasts` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.icmp_echo_ignore_broadcasts` runtime value is 1.
 - `net.ipv4.icmp_echo_ignore_broadcasts` is not persisted.
- **Recommendation:** Add `net.ipv4.icmp_echo_ignore_broadcasts = 1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.8: Ensure `net.ipv4.conf.all.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.all.accept_redirects` runtime value is 1. Expected is 0.

- `net.ipv4.conf.all.accept_redirects` is not persisted.

- **Recommendation:** Set `net.ipv4.conf.all.accept_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.9: Ensure `net.ipv4.conf.default.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:**

- `net.ipv4.conf.default.accept_redirects` runtime value is 1. Expected is 0.
- `net.ipv4.conf.default.accept_redirects` is not persisted.

- **Recommendation:** Set `net.ipv4.conf.default.accept_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.10: Ensure `net.ipv4.conf.all.secure_redirects` is configured

- **Status:** FAIL
- **Evidence:**

- `net.ipv4.conf.all.secure_redirects` runtime value is 1. Expected is 0.
- `net.ipv4.conf.all.secure_redirects` is not persisted.

- **Recommendation:** Set `net.ipv4.conf.all.secure_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.11: Ensure `net.ipv4.conf.default.secure_redirects` is configured

- **Status:** FAIL
- **Evidence:**

- `net.ipv4.conf.default.secure_redirects` runtime value is 1. Expected is 0.
- `net.ipv4.conf.default.secure_redirects` is not persisted.

- **Recommendation:** Set `net.ipv4.conf.default.secure_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.12: Ensure `net.ipv4.conf.all.rp_filter` is configured

- **Status:** PASS
- **Evidence:**

- `net.ipv4.conf.all.rp_filter` runtime value is 2.
- `net.ipv4.conf.all.rp_filter` is persisted with value 2 in `/etc/sysctl.d/10-network-security.conf`.

- **Recommendation:** None

Rule 3.3.1.13: Ensure net.ipv4.conf.default.rp_filter is configured

- **Status:** PASS
- **Evidence:**
 - net.ipv4.conf.default.rp_filter runtime value is 2.
 - net.ipv4.conf.default.rp_filter is persisted with value 2 in /etc/sysctl.d/10-network-security.conf.
- **Recommendation:** None

Rule 3.3.1.14: Ensure net.ipv4.conf.all.accept_source_route is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.all.accept_source_route runtime value is 0.
 - net.ipv4.conf.all.accept_source_route is not persisted.
- **Recommendation:** Add net.ipv4.conf.all.accept_source_route = 0 to a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.1.15: Ensure net.ipv4.conf.default.accept_source_route is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.default.accept_source_route runtime value is 1. Expected is 0.
 - net.ipv4.conf.default.accept_source_route is not persisted.
- **Recommendation:** Set net.ipv4.conf.default.accept_source_route = 0 in a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.1.16: Ensure net.ipv4.conf.all.log_martians is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.all.log_martians runtime value is 0. Expected is 1.
 - net.ipv4.conf.all.log_martians is not persisted.
- **Recommendation:** Set net.ipv4.conf.all.log_martians = 1 in a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.1.17: Ensure net.ipv4.conf.default.log_martians is configured

- **Status:** FAIL
- **Evidence:**

- `net.ipv4.conf.default.log_martians` runtime value is 0. Expected is 1.
- `net.ipv4.conf.default.log_martians` is not persisted.
- **Recommendation:** Set `net.ipv4.conf.default.log_martians = 1` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.18: Ensure `net.ipv4.tcp_syncookies` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.tcp_syncookies` runtime value is 1.
 - `net.ipv4.tcp_syncookies` is not persisted.
- **Recommendation:** Add `net.ipv4.tcp_syncookies = 1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.1: Ensure `net.ipv6.conf.all.forwarding` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.all.forwarding` runtime value is 0.
 - `net.ipv6.conf.all.forwarding` is not persisted.
- **Recommendation:** Add `net.ipv6.conf.all.forwarding = 0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.2: Ensure `net.ipv6.conf.default.forwarding` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.default.forwarding` runtime value is 0.
 - `net.ipv6.conf.default.forwarding` is not persisted.
- **Recommendation:** Add `net.ipv6.conf.default.forwarding = 0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.3: Ensure `net.ipv6.conf.all.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.all.accept_redirects` runtime value is 1. Expected is 0.
 - `net.ipv6.conf.all.accept_redirects` is not persisted.
- **Recommendation:** Set `net.ipv6.conf.all.accept_redirects = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.4: Ensure net.ipv6.conf.default.accept_redirects is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv6.conf.default.accept_redirects runtime value is 1. Expected is 0.
 - net.ipv6.conf.default.accept_redirects is not persisted.
- **Recommendation:** Set net.ipv6.conf.default.accept_redirects = 0 in a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.2.5: Ensure net.ipv6.conf.all.accept_source_route is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv6.conf.all.accept_source_route runtime value is 0.
 - net.ipv6.conf.all.accept_source_route is not persisted.
- **Recommendation:** Add net.ipv6.conf.all.accept_source_route = 0 to a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.2.6: Ensure net.ipv6.conf.default.accept_source_route is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv6.conf.default.accept_source_route runtime value is 0.
 - net.ipv6.conf.default.accept_source_route is not persisted.
- **Recommendation:** Add net.ipv6.conf.default.accept_source_route = 0 to a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.2.7: Ensure net.ipv6.conf.all.accept_ra is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv6.conf.all.accept_ra runtime value is 1. Expected is 0.
 - net.ipv6.conf.all.accept_ra is not persisted.
- **Recommendation:** Set net.ipv6.conf.all.accept_ra = 0 in a sysctl configuration file (e.g., /etc/sysctl.d/99-sysctl.conf) and apply with `sudo sysctl -p`.

Rule 3.3.2.8: Ensure net.ipv6.conf.default.accept_ra is configured

- **Status:** FAIL

- **Evidence:**
 - `net.ipv6.conf.default.accept_ra` runtime value is 1. Expected is 0.
 - `net.ipv6.conf.default.accept_ra` is not persisted.
- **Recommendation:** Set `net.ipv6.conf.default.accept_ra = 0` in a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Section 4.1 – Configure Uncomplicated Firewall

Rule 4.1.1: Ensure ufw is installed

- **Status:** PASS
- **Evidence:** `ufw` package is installed (`ufw_installed` is `true`).
- **Recommendation:** None

Rule 4.1.2: Ensure ufw service is configured

- **Status:** FAIL
- **Evidence:** `ufw` service is enabled but `Status: inactive`. This indicates the service is not actively enforcing firewall rules.
- **Recommendation:** Enable and start the UFW firewall service using `sudo ufw enable`.

Rule 4.1.3: Ensure ufw incoming default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_INPUT_POLICY` in `/etc/default/ufw` is set to `DROP`.
- **Recommendation:** None

Rule 4.1.4: Ensure ufw outgoing default is configured

- **Status:** FAIL
- **Evidence:** `DEFAULT_OUTPUT_POLICY` in `/etc/default/ufw` is set to `ACCEPT`.
- **Recommendation:** Change `DEFAULT_OUTPUT_POLICY` to `DROP` in `/etc/default/ufw` and configure explicit allow rules for necessary outbound traffic.

Rule 4.1.5: Ensure ufw routed default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_FORWARD_POLICY` in `/etc/default/ufw` is set to `DROP`.
- **Recommendation:** None

Section 5.3 – Pluggable Authentication Modules (PAM)

Rule 5.3.1.1: Ensure latest version of pam is installed

- **Status:** PASS
- **Evidence:** `libpam-runtime` is installed (version `1.5.3-5ubuntu5.5`) and is the candidate version.

- **Recommendation:** None

Rule 5.3.1.2: Ensure latest version of libpam-modules is installed

- **Status:** PASS
- **Evidence:** libpam-modules is installed (version 1.5.3-5ubuntu5.5) and is the candidate version.
- **Recommendation:** None

Rule 5.3.1.3: Ensure latest version of libpam-pwquality is installed

- **Status:** FAIL
- **Evidence:** libpam-pwquality package is not installed.
- **Recommendation:** Install libpam-pwquality: `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure latest version of cracklib-runtime is installed

- **Status:** FAIL
- **Evidence:** cracklib-runtime package is not installed.
- **Recommendation:** Install cracklib-runtime: `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure pam_unix module is enabled

- **Status:** PASS
- **Evidence:** pam_unix.so module is enabled in common-auth, common-account, common-password, and common-session.
- **Recommendation:** None

Rule 5.3.2.2: Ensure pam_faillock module is enabled

- **Status:** FAIL
- **Evidence:** pam_faillock.so module is not enabled in any PAM configuration files.
- **Recommendation:** Enable pam_faillock.so in /etc/pam.d/common-auth and /etc/pam.d/common-account to enforce account lockout policies.

Rule 5.3.2.3: Ensure pam_pwquality module is enabled

- **Status:** FAIL
- **Evidence:** pam_pwquality.so module is not enabled in any PAM configuration files.
- **Recommendation:** Enable pam_pwquality.so in /etc/pam.d/common-password to enforce password quality requirements.

Rule 5.3.2.4: Ensure pam_pwhistory module is enabled

- **Status:** FAIL
- **Evidence:** pam_pwhistory.so module is not enabled in any PAM configuration files.
- **Recommendation:** Enable pam_pwhistory.so in /etc/pam.d/common-password to enforce password history.

Rule 5.3.3.1.1: Ensure password failed attempts lockout is configured

- **Status:** FAIL
- **Evidence:** deny setting for pam_faillock is not configured in /etc/security/faillock.conf or inline PAM arguments.
- **Recommendation:** Configure deny = 5 (or less) in /etc/security/faillock.conf and ensure pam_faillock.so is enabled in PAM.

Rule 5.3.3.1.2: Ensure password unlock time is configured

- **Status:** FAIL
- **Evidence:** unlock_time setting for pam_faillock is not configured in /etc/security/faillock.conf or inline PAM arguments.
- **Recommendation:** Configure unlock_time = 900 (or 0 for permanent lockout) in /etc/security/faillock.conf and ensure pam_faillock.so is enabled in PAM.

Rule 5.3.3.1.3: Ensure password failed attempts lockout includes root account

- **Status:** FAIL
- **Evidence:** even_deny_root setting for pam_faillock is not configured in /etc/security/faillock.conf or inline PAM arguments.
- **Recommendation:** Add even_deny_root to /etc/security/faillock.conf and ensure pam_faillock.so is enabled in PAM.

Rule 5.3.3.2.1: Ensure password number of changed characters is configured

- **Status:** FAIL
- **Evidence:** difok setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure difok = 2 (or more) in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.2.2: Ensure password length is configured

- **Status:** FAIL
- **Evidence:** minlen setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure minlen = 14 (or more) in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.2.3: Ensure password complexity is configured

- **Status:** INFORMATIONAL
- **Evidence:** dcredit, ucredit, lcredit, ocredit settings for pam_pwquality are not explicitly configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Manually review and configure dcredit, ucredit, lcredit, ocredit in /etc/security/pwquality.conf to enforce specific character type requirements (e.g., dcredit=-1, ucredit=-1, lcredit=-1, ocredit=-1).

Rule 5.3.3.2.4: Ensure password same consecutive characters is

configured

- **Status:** FAIL
- **Evidence:** maxrepeat setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure maxrepeat = 3 (or less) in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.2.5: Ensure password maximum sequential characters is configured

- **Status:** FAIL
- **Evidence:** maxsequence setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure maxsequence = 3 (or less) in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.2.6: Ensure password dictionary check is enabled

- **Status:** FAIL
- **Evidence:** dictcheck setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure dictcheck = 1 in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.2.7: Ensure password quality checking is enforced

- **Status:** FAIL
- **Evidence:** enforcing setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure enforcing = 1 in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.2.8: Ensure password quality is enforced for the root user

- **Status:** FAIL
- **Evidence:** enforce_for_root setting for pam_pwquality is not configured in /etc/security/pwquality.conf or inline PAM arguments.
- **Recommendation:** Configure enforce_for_root in /etc/security/pwquality.conf and ensure pam_pwquality.so is enabled in PAM.

Rule 5.3.3.3.1: Ensure password history remember is configured

- **Status:** FAIL
- **Evidence:** remember setting for pam_pwhistory is not configured in inline PAM arguments.
- **Recommendation:** Configure remember = 24 (or more) in the pam_pwhistory.so line in /etc/pam.d/common-password and ensure pam_pwhistory.so is enabled.

Rule 5.3.3.3.2: Ensure password history is enforced for the root user

- **Status:** FAIL

- **Evidence:** `enforce_for_root` setting for `pam_pwhistory` is not configured in inline PAM arguments.
- **Recommendation:** Add `enforce_for_root` to the `pam_pwhistory.so` line in `/etc/pam.d/common-password` and ensure `pam_pwhistory.so` is enabled.

Rule 5.3.3.3.3: Ensure `pam_pwhistory` includes `use_authtok`

- **Status:** FAIL
- **Evidence:** `use_authtok` setting for `pam_pwhistory` is not configured in inline PAM arguments.
- **Recommendation:** Add `use_authtok` to the `pam_pwhistory.so` line in `/etc/pam.d/common-password` and ensure `pam_pwhistory.so` is enabled.

Rule 5.3.3.4.1: Ensure `pam_unix` does not include `nullok`

- **Status:** FAIL
- **Evidence:** The `pam_unix.so` line in `/etc/pam.d/common-auth` includes the `nullok` option.
- **Recommendation:** Remove the `nullok` option from the `pam_unix.so` line in `/etc/pam.d/common-auth`.

Rule 5.3.3.4.2: Ensure `pam_unix` does not include `remember`

- **Status:** PASS
- **Evidence:** The `pam_unix.so` lines do not include the `remember` option.
- **Recommendation:** None

Rule 5.3.3.4.3: Ensure `pam_unix` includes a strong password hashing algorithm

- **Status:** PASS
- **Evidence:** The `pam_unix.so` line in `/etc/pam.d/common-password` includes the `yescrypt` option.
- **Recommendation:** None

Rule 5.3.3.4.4: Ensure `pam_unix` includes `use_authtok`

- **Status:** FAIL
- **Evidence:** The `pam_unix.so` line in `/etc/pam.d/common-password` does not explicitly include the `use_authtok` option.
- **Recommendation:** Add the `use_authtok` option to the `pam_unix.so` line in `/etc/pam.d/common-password`.

Section 5.4 – User Accounts and Environment

Rule 5.4.1.1: Ensure password expiration is configured

- **Status:** FAIL
- **Evidence:** `PASS_MAX_DAYS` in `/etc/login.defs` is set to 99999.
- **Recommendation:** Set `PASS_MAX_DAYS` to 90 (or less) in `/etc/login.defs` to enforce regular password changes.

Rule 5.4.1.2: Ensure minimum password days is configured

- **Status:** INFORMATIONAL
- **Evidence:** PASS_MIN_DAYS in /etc/login.defs is set to 0.
- **Recommendation:** Consider setting PASS_MIN_DAYS to 1 or more in /etc/login.defs to prevent immediate password changes.

Rule 5.4.1.3: Ensure password expiration warning days is configured

- **Status:** PASS
- **Evidence:** PASS_WARN_AGE in /etc/login.defs is set to 7.
- **Recommendation:** None

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured

- **Status:** PASS
- **Evidence:** ENCRYPT_METHOD in /etc/login.defs is SHA512, and pam_unix is configured to use yescrypt.
- **Recommendation:** None

Rule 5.4.1.5: Ensure inactive password lock is configured

- **Status:** FAIL
- **Evidence:** useradd_inactive is set to -1 (never inactive).
- **Recommendation:** Set INACTIVE to 30 (or less) in /etc/default/useradd to lock accounts after a period of inactivity.

Rule 5.4.1.6: Ensure all users last password change date is in the past

- **Status:** FAIL
- **Evidence:** The last_password_change date for several users (e.g., root, daemon, bin, sys, sync, games, man, lp, mail, news, uucp, proxy, www-data, backup, list, irc, _apt, nobody, systemd-network, systemd-timesync, dhcpcd, messagebus, systemd-resolve, pollinate, polkitd, syslog, uidd, tcpdump, tss, landscape, fwupd-refresh) is Feb 16, 2025, Apr 19, 2026, or Jun 25, 2026, which are in the future.
- **Recommendation:** Force a password change for all users with future password change dates using `sudo chage -d 0 <username>`.

Rule 5.4.2.1: Ensure root is the only UID 0 account

- **Status:** PASS
- **Evidence:** Only root is listed as a UID 0 account.
- **Recommendation:** None

Rule 5.4.2.2: Ensure root is the only GID 0 account

- **Status:** PASS
- **Evidence:** Only root is listed as a primary GID 0 account.
- **Recommendation:** None

Rule 5.4.2.3: Ensure group root is the only GID 0 group

- **Status:** PASS
- **Evidence:** Only `root` is listed as a GID 0 group.
- **Recommendation:** None

Rule 5.4.2.4: Ensure root account access is controlled

- **Status:** FAIL
- **Evidence:** `securetty_exists` is false, indicating that `/etc/securetty` is not used to restrict root logins.
- **Recommendation:** Create and configure `/etc/securetty` to list only allowed TTYs for root login.

Rule 5.4.2.5: Ensure root path integrity

- **Status:** PASS
- **Evidence:** The root path is `/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin`. This path does not contain any dot-directories or world-writable components.
- **Recommendation:** None

Rule 5.4.2.6: Ensure root user umask is configured

- **Status:** FAIL
- **Evidence:** The umask for root is 022 (from `/etc/login.defs`).
- **Recommendation:** Set the root user's umask to 027 or 077 in `/etc/profile`, `/etc/bash.bashrc`, or `/etc/login.defs`.

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell

- **Status:** FAIL
- **Evidence:** The `sync` system account has `/bin/sync` as its login shell, which is a valid shell.
- **Recommendation:** Change the shell for the `sync` user to `/usr/sbin/nologin` or `/bin/false` using `sudo usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked

- **Status:** PASS
- **Evidence:** All system accounts with non-login shells have a `passwd_status` of L (locked).
- **Recommendation:** None

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells

- **Status:** PASS
- **Evidence:** `nologin_in_shells` is false.
- **Recommendation:** None

Rule 5.4.3.2: Ensure default user shell timeout is configured

- **Status:** FAIL
- **Evidence:** No `TMOUT` settings were found in `/etc/profile`, `/etc/profile.d/`, or `/`

`etc/bash.bashrc`.

- **Recommendation:** Add `TMOUT=600` (or less) to `/etc/profile.d/timeout.sh` to automatically log out inactive shell sessions.

Rule 5.4.3.3: Ensure default user umask is configured

- **Status:** FAIL
- **Evidence:** The default user umask is `022` (from `/etc/login.defs`).
- **Recommendation:** Set the default user umask to `027` or `077` in `/etc/login.defs` or via `pam_umask.so` configuration.

Section 6.1 – System Logging

Rule 6.1.1.1.1: Ensure journald service is active

- **Status:** PASS
- **Evidence:** `systemd-journald.service` is active.
- **Recommendation:** None

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use

- **Status:** PASS
- **Evidence:** `systemd-journal-remote` package is not installed, and `systemd-journal-remote.socket` is inactive.
- **Recommendation:** None

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog

- **Status:** FAIL
- **Evidence:** `ForwardToSyslog=no` is commented out in `/etc/systemd/journald.conf`, meaning it's not explicitly configured to forward logs to rsyslog.
- **Recommendation:** Uncomment and set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` and restart `systemd-journald`.

Rule 6.1.1.1.4: Ensure journald log file access is configured

- **Status:** INFORMATIONAL
- **Evidence:**
 - `/run/log/journal` has mode `2755`, owner `root`, group `999`.
 - `/var/log/journal` has mode `2755`, owner `root`, group `999`.
 - Log files within `/var/log/journal` have mode `0640`, owner `root`, group `999`.
- **Recommendation:** Manually verify that the permissions and ownership of journald log files and directories are appropriate for the environment and meet security requirements.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured

- **Status:** INFORMATIONAL
- **Evidence:** Log rotation settings (`SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, `MaxFileSec`) are

commented out in `/etc/systemd/journald.conf`, indicating default values are in use.

- **Recommendation:** Explicitly configure log rotation settings in `/etc/systemd/journald.conf` (e.g., `SystemMaxUse=1G`, `MaxRetentionSec=1month`) to ensure logs are managed effectively.

Rule 6.1.1.1.6: Ensure journald Storage is configured

- **Status:** FAIL
- **Evidence:** `Storage=auto` is commented out in `/etc/systemd/journald.conf`, meaning it's not explicitly configured for persistent storage.
- **Recommendation:** Uncomment and set `Storage=persistent` in `/etc/systemd/journald.conf` and restart `systemd-journald`.

Rule 6.1.1.1.7: Ensure journald Compress is configured

- **Status:** FAIL
- **Evidence:** `Compress=yes` is commented out in `/etc/systemd/journald.conf`, meaning it's not explicitly configured to compress logs.
- **Recommendation:** Uncomment and set `Compress=yes` in `/etc/systemd/journald.conf` and restart `systemd-journald`.

Rule 6.1.2.1: Ensure rsyslog is installed

- **Status:** PASS
- **Evidence:** `rsyslog` package is installed.
- **Recommendation:** None

Rule 6.1.2.2: Ensure rsyslog service is enabled and active

- **Status:** PASS
- **Evidence:** `rsyslog.service` is enabled and active.
- **Recommendation:** None

Rule 6.1.2.3: Ensure rsyslog log file creation mode is configured

- **Status:** PASS
- **Evidence:** `/etc/rsyslog.conf` contains `$FileCreateMode 0640`.
- **Recommendation:** None

Rule 6.1.2.4: Ensure rsyslog logging is configured

- **Status:** INFORMATIONAL
- **Evidence:** `rsyslog` configuration files (`/etc/rsyslog.conf`, `/etc/rsyslog.d/20-ufw.conf`, `/etc/rsyslog.d/21-cloudinit.conf`, `/etc/rsyslog.d/50-default.conf`) define various logging rules.
- **Recommendation:** Manually review the `rsyslog` configuration to ensure all necessary system events are logged to appropriate files.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host

- **Status:** INFORMATIONAL
- **Evidence:** `rsyslog` configuration files do not explicitly show remote log forwarding rules.
- **Recommendation:** Configure `rsyslog` to send logs to a central, secure log management server.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client

- **Status:** PASS
- **Evidence:** `module(load="imudp")` and `module(load="imtcp")` are commented out in `/etc/rsyslog.conf`, indicating `rsyslog` is not configured to receive logs from remote clients.
- **Recommendation:** None

Rule 6.1.2.7: Ensure logrotate is configured

- **Status:** INFORMATIONAL
- **Evidence:** `logrotate` is configured via `/etc/logrotate.conf` and includes files from `/etc/logrotate.d/`. The `logrotate.timer` is active and enabled.
- **Recommendation:** Manually review `logrotate` configurations for all critical log files to ensure appropriate rotation, compression, and retention policies are in place.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed

- **Status:** FAIL
- **Evidence:** `rsyslog-gnutls` package is not installed.
- **Recommendation:** Install `rsyslog-gnutls` to enable encrypted remote log forwarding: `sudo apt install rsyslog-gnutls`.

Rule 6.1.2.9: Ensure rsyslog forwarding uses gtls

- **Status:** FAIL
- **Evidence:** `rsyslog-gnutls` is not installed, so `gtls` cannot be used for forwarding.
- **Recommendation:** Install `rsyslog-gnutls` and configure remote log forwarding to use `gtls` for encryption.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured

- **Status:** INFORMATIONAL
- **Evidence:** `rsyslog-gnutls` is not installed, so CA certificates for `gtls` are not configured.
- **Recommendation:** If remote log forwarding is implemented with `gtls`, ensure CA certificates are properly configured for secure communication.

Rule 6.1.3.1: Ensure access to all logfiles has been configured

- **Status:** FAIL
- **Evidence:** Several log files have overly permissive permissions:
 - `/var/log/wtmp` has mode `0664`.
 - `/var/log/btmp` has mode `0660`.

- /var/log/lastlog has mode 0664.
- /var/log/faillog has mode 0644.
- /var/log/dpkg.log has mode 0644.
- /var/log/alternatives.log has mode 0644.
- /var/log/apt/history.log has mode 0644.
- /var/log/unattended-upgrades/unattended-upgrades.log has mode 0644.
- /var/log/unattended-upgrades/unattended-upgrades-shutdown.log has mode 0644.
- /var/log/landscape/sysinfo.log has mode 0644.
- /var/log/sysstat/sa* and sar* files have mode 0644.
- /var/log/wazuh-indexer/gc.log has mode 0644.
- /var/log/wazuh-indexer/wazuh-cluster_server.json has mode 0644.

- **Recommendation:** Restrict permissions on all log files to 0640 or 0600 to prevent unauthorized access. For example, `sudo chmod 0640 /var/log/wtmp`.

Section 6.2 – System Auditing

Rule 6.2.1.1: Ensure auditd packages are installed

- **Status:** FAIL
- **Evidence:** auditd package is not installed.
- **Recommendation:** Install auditd and audispd-plugins: `sudo apt install auditd audispd-plugins`.

Rule 6.2.1.2: Ensure auditd service is enabled and active

- **Status:** NOT APPLICABLE
- **Evidence:** auditd is not installed.
- **Recommendation:** Install and enable auditd service.

Rule 6.2.1.3: Ensure auditing for processes that start prior to auditd is enabled

- **Status:** UNKNOWN
- **Evidence:** Information about GRUB kernel parameters for `audit=1` is not available in the profile.
- **Recommendation:** If auditd is installed, ensure `audit=1` is added to the `GRUB_CMDLINE_LINUX` variable in `/etc/default/grub`, then run `sudo update-grub` and reboot.

Rule 6.2.1.4: Ensure audit_backlog_limit is configured

- **Status:** UNKNOWN
- **Evidence:** Information about `audit_backlog_limit` in `auditd.conf` is not available in the profile.
- **Recommendation:** If auditd is installed, configure `backlog = 8192` (or higher) in `/etc/audit/auditd.conf`.

Rule 6.2.2.1: Ensure audit log storage size is configured

- **Status:** UNKNOWN
- **Evidence:** Information about `max_log_file` in `auditd.conf` is not available in the profile.
- **Recommendation:** If `auditd` is installed, configure `max_log_file` in `/etc/audit/auditd.conf` to an appropriate size (e.g., `max_log_file = 50`).

Rule 6.2.2.2: Ensure audit logs are not automatically deleted

- **Status:** UNKNOWN
- **Evidence:** Information about `max_log_file_action` in `auditd.conf` is not available in the profile.
- **Recommendation:** If `auditd` is installed, configure `max_log_file_action = keep_logs` in `/etc/audit/auditd.conf`.

Rule 6.2.2.3: Ensure system is disabled when audit logs are full

- **Status:** UNKNOWN
- **Evidence:** Information about `disk_full_action` in `auditd.conf` is not available in the profile.
- **Recommendation:** If `auditd` is installed, configure `disk_full_action = halt` or `disk_full_action = single` in `/etc/audit/auditd.conf`.

Rule 6.2.2.4: Ensure system warns when audit logs are low on space

- **Status:** UNKNOWN
- **Evidence:** Information about `space_left_action` in `auditd.conf` is not available in the profile.
- **Recommendation:** If `auditd` is installed, configure `space_left_action = SYSLOG` or `space_left_action = EMAIL` in `/etc/audit/auditd.conf`.

Rule 6.2.3.1: Ensure changes to system administration scope (sudoers) is collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit changes to `/etc/sudoers` and `/etc/sudoers.d/`.

Rule 6.2.3.2: Ensure actions as another user are always logged

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit actions performed as another user.

Rule 6.2.3.3: Ensure events that modify the sudo log file are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to the sudo log file.

Rule 6.2.3.4: Ensure events that modify date and time information are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit changes to date/time and `/etc/localtime`.

Rule 6.2.3.5: Ensure events that modify `sethostname` and `setdomainname` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit `sethostname` and `setdomainname` system calls.

Rule 6.2.3.6: Ensure events that modify `/etc/issue` and `/etc/issue.net` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/issue` and `/etc/issue.net`.

Rule 6.2.3.7: Ensure events that modify `/etc/hosts` and `/etc/hostname` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/hosts` and `/etc/hostname`.

Rule 6.2.3.8: Ensure events that modify `/etc/network` and `/etc/networks` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/network` and `/etc/networks`.

Rule 6.2.3.9: Ensure events that modify `/etc/netplan` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/netplan`.

Rule 6.2.3.10: Ensure use of privileged commands are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.

- **Recommendation:** Install `auditd` and add rules to audit the execution of privileged commands.

Rule 6.2.3.11: Ensure events that modify `/etc/group` information are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/group`.

Rule 6.2.3.12: Ensure events that modify `/etc/passwd` information are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/passwd`.

Rule 6.2.3.13: Ensure events that modify `/etc/shadow` and `/etc/gshadow` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/shadow` and `/etc/gshadow`.

Rule 6.2.3.14: Ensure events that modify `/etc/security/opasswd` are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/security/opasswd`.

Rule 6.2.3.15: Ensure events that modify `/etc/nsswitch.conf` file are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to `/etc/nsswitch.conf`.

Rule 6.2.3.16: Ensure events that modify `/etc/pam.conf` and `/etc/pam.d/` information are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to PAM configuration files.

Rule 6.2.3.17: Ensure unsuccessful file access attempts are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit unsuccessful file access attempts.

Rule 6.2.3.18: Ensure discretionary access control permission modification events are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit DAC permission modification events.

Rule 6.2.3.19: Ensure successful file system mounts are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit successful file system mounts.

Rule 6.2.3.20: Ensure session initiation information is collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit session initiation events.

Rule 6.2.3.21: Ensure login and logout events are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit login and logout events.

Rule 6.2.3.22: Ensure file deletion events by users are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit file deletion events.

Rule 6.2.3.23: Ensure events that modify the system's Mandatory Access Controls are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit modifications to MAC policies.

Rule 6.2.3.24: Ensure successful and unsuccessful attempts to use the `chcon` command are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit `chcon` command usage.

Rule 6.2.3.25: Ensure successful and unsuccessful attempts to use the `setfacl` command are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit `setfacl` command usage.

Rule 6.2.3.26: Ensure successful and unsuccessful attempts to use the `chacl` command are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit `chacl` command usage.

Rule 6.2.3.27: Ensure successful and unsuccessful attempts to use the `usermod` command are collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit `usermod` command usage.

Rule 6.2.3.28: Ensure kernel module loading unloading and modification is collected

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add rules to audit kernel module events.

Rule 6.2.3.29: Ensure the audit configuration is immutable

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed, so audit rules are not configured.
- **Recommendation:** Install `auditd` and add the `-e 2` rule to `/etc/audit/rules.d/audit.rules` to make the audit configuration immutable.

Rule 6.2.3.30: Ensure the running and on disk configuration is the same

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure the running and on-disk configurations are synchronized.

Rule 6.2.4.1: Ensure the audit log file directory mode is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure `/var/log/audit` has mode `0750`.

Rule 6.2.4.2: Ensure audit log files mode is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit log files have mode `0640`.

Rule 6.2.4.3: Ensure audit log files owner is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit log files are owned by `root`.

Rule 6.2.4.4: Ensure audit log files group owner is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit log files are group-owned by `adm`.

Rule 6.2.4.5: Ensure audit configuration files mode is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit configuration files have mode `0640`.

Rule 6.2.4.6: Ensure audit configuration files owner is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit configuration files are owned by `root`.

Rule 6.2.4.7: Ensure audit configuration files group owner is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit configuration files are group-owned by `root`.

Rule 6.2.4.8: Ensure audit tools mode is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit tools have mode `0755`.

Rule 6.2.4.9: Ensure audit tools owner is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit tools are owned by `root`.

Rule 6.2.4.10: Ensure audit tools group owner is configured

- **Status:** UNKNOWN
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and ensure audit tools are group-owned by `root`.

Section 6.3 – Configure Integrity Checking

Rule 6.3.1: Ensure AIDE is installed

- **Status:** FAIL
- **Evidence:** `aide_installed` is false.
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`.

Rule 6.3.2: Ensure filesystem integrity is regularly checked

- **Status:** FAIL
- **Evidence:** `scheduled_checking` is null, indicating no scheduled AIDE checks are configured.
- **Recommendation:** Install AIDE and configure a cron job or systemd timer to regularly run AIDE checks (e.g., daily or weekly).

Rule 6.3.3: Ensure cryptographic mechanisms are used to protect the integrity of audit tools

- **Status:** FAIL
- **Evidence:** `audit_tools_integrity_tracked` is null, indicating AIDE is not configured to track the integrity of audit tools.
- **Recommendation:** Install AIDE and configure its database to include audit tools with strong cryptographic hashes (e.g., `sha512`).

Section 7.1 – Configure system file and directory access

Rule 7.1.1: Ensure access to `/etc/passwd` is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd` has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.2: Ensure access to `/etc/passwd-` is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd-` has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.3: Ensure access to `/etc/group` is configured

- **Status:** PASS
- **Evidence:** `/etc/group` has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.4: Ensure access to `/etc/group-` is configured

- **Status:** PASS
- **Evidence:** `/etc/group` has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.5: Ensure access to `/etc/shadow` is configured

- **Status:** PASS
- **Evidence:** `/etc/shadow` has mode `0o640`, owner `root`, group `shadow`.
- **Recommendation:** None

Rule 7.1.6: Ensure access to `/etc/shadow-` is configured

- **Status:** PASS
- **Evidence:** `/etc/shadow-` has mode `0o640`, owner `root`, group `shadow`.
- **Recommendation:** None

Rule 7.1.7: Ensure access to `/etc/gshadow` is configured

- **Status:** PASS
- **Evidence:** `/etc/gshadow` has mode `0o640`, owner `root`, group `shadow`.
- **Recommendation:** None

Rule 7.1.8: Ensure access to `/etc/gshadow-` is configured

- **Status:** PASS
- **Evidence:** `/etc/gshadow-` has mode `0o640`, owner `root`, group `shadow`.
- **Recommendation:** None

Rule 7.1.9: Ensure access to `/etc/shells` is configured

- **Status:** PASS
- **Evidence:** `/etc/shells` has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.10: Ensure access to `/etc/security/opasswd` is configured

- **Status:** PASS
- **Evidence:** `/etc/security/opasswd` has mode `0o600`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.11: Ensure world writable files and directories are secured

- **Status:** PASS
- **Evidence:** No world-writable files or directories without the sticky bit were found in the scanned mounts (`/`, `/boot`).
- **Recommendation:** None

Rule 7.1.12: Ensure no files or directories without an owner and a group exist

- **Status:** PASS
- **Evidence:** No unowned or ungrouped files or directories were found in the scanned

mounts (/ , /boot).

- **Recommendation:** None

Rule 7.1.13: Ensure SUID and SGID files are reviewed

- **Status:** INFORMATIONAL
- **Evidence:** The following SUID/SGID files were found:
 - /usr/bin/umount (4755 root:root)
 - /usr/bin/chsh (4755 root:root)
 - /usr/bin/sudo (4755 root:root)
 - /usr/bin/su (4755 root:root)
 - /usr/bin/passwd (4755 root:root)
 - /usr/bin/ssh-agent (2755 root:_ssh)
 - /usr/bin/fusermount3 (4755 root:root)
 - /usr/bin/newgrp (4755 root:root)
 - /usr/bin/chage (2755 root:shadow)
 - /usr/bin/expiry (2755 root:shadow)
 - /usr/bin/gpasswd (4755 root:root)
 - /usr/bin/crontab (2755 root:crontab)
 - /usr/bin/mount (4755 root:root)
 - /usr/bin/chfn (4755 root:root)
 - /usr/lib/dbus-1.0/dbus-daemon-launch-helper (4754 root:messagebus)
 - /usr/lib/x86_64-linux-gnu/utempter/utempter (2755 root:utmp)
 - /usr/lib/openssh/ssh-keysign (4755 root:root)
 - /usr/lib/polkit-1/polkit-agent-helper-1 (4755 root:root)
 - /usr/sbin/pam_extrausers_chkpwd (2755 root:shadow)
 - /usr/sbin/unix_chkpwd (2755 root:shadow)
- **Recommendation:** Periodically review the list of SUID/SGID files to ensure that only authorized and necessary programs have these permissions. Remove SUID/SGID bits from any files that do not require them.

Section 7.2 – Local User and Group Settings

Rule 7.2.1: Ensure accounts in /etc/passwd use shadowed passwords

- **Status:** PASS
- **Evidence:** /etc/shadow exists and is configured with appropriate permissions (7.1.5), indicating shadowed passwords are in use.
- **Recommendation:** None

Rule 7.2.2: Ensure /etc/shadow password fields are not empty

- **Status:** PASS
- **Evidence:** The root account's password status is *, indicating a password is set (not empty).
- **Recommendation:** None

Rule 7.2.3: Ensure all groups in /etc/passwd exist in /etc/group

- **Status:** PASS
- **Evidence:** `orphan_gids_in_passwd` is empty, indicating no orphan GIDs were found.
- **Recommendation:** None

Rule 7.2.4: Ensure shadow group is empty

- **Status:** PASS
- **Evidence:** `shadow_group_members` is empty, indicating no users are members of the shadow group.
- **Recommendation:** None

Rule 7.2.5: Ensure no duplicate UIDs exist

- **Status:** PASS
- **Evidence:** `duplicate_uids` is empty, indicating no duplicate UIDs were found.
- **Recommendation:** None

Rule 7.2.6: Ensure no duplicate GIDs exist

- **Status:** PASS
- **Evidence:** `duplicate_gids` is empty, indicating no duplicate GIDs were found.
- **Recommendation:** None

Rule 7.2.7: Ensure no duplicate user names exist

- **Status:** PASS
- **Evidence:** `duplicate_usernames` is empty, indicating no duplicate user names were found.
- **Recommendation:** None

Rule 7.2.8: Ensure no duplicate group names exist

- **Status:** PASS
- **Evidence:** `duplicate_groupnames` is empty, indicating no duplicate group names were found.
- **Recommendation:** None

Rule 7.2.9: Ensure local interactive user home directories are configured

- **Status:** PASS
- **Evidence:** `home_directory_issues` is empty, indicating no issues with home directory configuration.
- **Recommendation:** None

Rule 7.2.10: Ensure local interactive user dot files access is configured

- **Status:** PASS
- **Evidence:** `dotfile_permission_issues` is empty, indicating no issues with dot file permissions.
- **Recommendation:** None

SUID/SGID Triage Report

```
{
  "findings": [
    {
      "path": "/usr/bin/ssh-agent",
      "owner": "root",
      "group": "_ssh",
      "mode": "2755",
      "suid": false,
      "sgid": true,
      "risk": "medium",
      "reason": "The '/usr/bin/ssh-agent' binary typically does not req",
      "recommendation": "Investigate the necessity of the SGID bit on "
    }
  ],
  "summary": {
    "total_checked": 20,
    "suspicious_found": 1
  }
}
```